PARISHRAM 2.0

9:25 AM

CLASS 12TH 2026

Electrostatic Potential and Capacitance

PHYSICS | LECTURE-2

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Recap

1. Electrostatic potential → $V = \frac{kq}{r}$ → Scalar quantity.
2. Potential difference $\Delta V = V_2 - V_1 = \frac{W}{q}$
3. Electric potential due to
   1. A point charge → $\frac{kq}{r}$
   2. A dipole → $\frac{kp\cos\theta}{r^2}$
   3. System of charges → $V = V_1 + V_2 + V_3 \cdots$

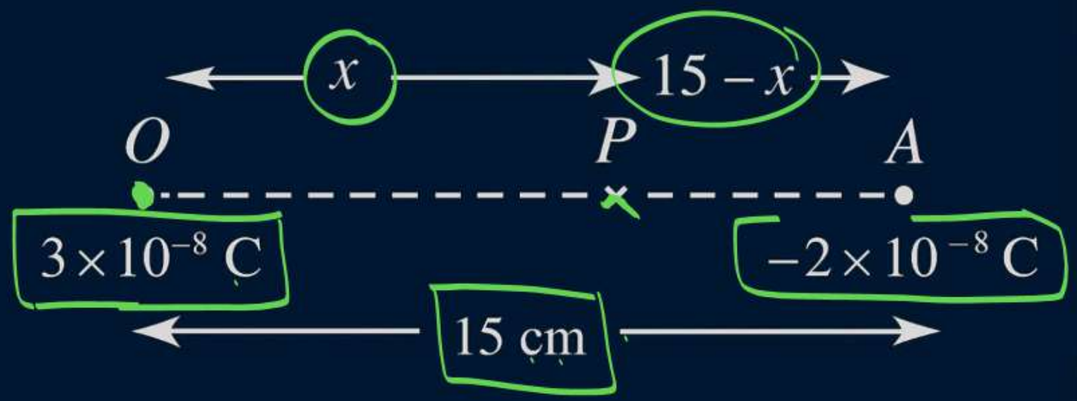


QUESTION

The point where Potential will be zero is at 9cm from the charge $3\times10^{-8}C$ on the line joining both charges.

Two charges 3×10^{-8}C and -2×10^{-8}C are located 15 cm apart. At what point on the line joining the two charges is the electric potential zero? Take the potential at infinity to be zero.

$$V_p = \frac{Kq_O}{x} + \frac{Kq_A}{(15-x)} = 0$$



$$\Rightarrow \frac{K(3\times10^{-8}\,C)}{x} + \frac{K(-2\times10^{-8}\,C)}{(15-x)} = 0$$

$$\Rightarrow \frac{3\times10^{-8}}{x} = \frac{2\times10^{-8}}{15-x} \Rightarrow 45-3x = 2x \Rightarrow 5x = 45 \Rightarrow \boxed{x = 9\,cm}$$

QUESTION

A point charge $Q = 5.0\mu C$ is fixed at the origin. What is the potential difference $V(0.5\ m) - V(0.2\ m)$? (Take $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9 Nm^2/C^2$.)

$$\Delta V = \frac{kQ}{0.5} - \frac{kQ}{0.2}$$

$$\Delta V = (9\times10^9)(5\times10^{-6})[2-5]$$

$$= 9\times5\times(-3)\times10^3$$

$$= -135\times10^3 = -1.35\times10^5\ V$$

A $+ 1.35 \times 10^5$ V

B $- 1.35 \times 10^5$ V

C $+ 4.50 \times 10^4$ V

D $- 4.50 \times 10^4$ V



QUESTION

$p = q(2a)$ $\Big\}$ $\rightarrow q = 2\,\mu C$, $2a = d = 4\ cm$

A dipole consists of charges $+2.0\mu C$ and $-2.0\mu C$ separated by 4.0 cm . Using the dipole approximation (point at distance r from dipole centre, $r \gg$ separation), the potential on the axial line is $V = \frac{kp}{r^2}$ with $p = qd$. Find V at a point on the axial line at $r = 10.0$ cm from the dipole centre. (Use $k = 9.0 \times 10^9$.)

A $+ 7.2 \times 10^4$ V

$V = \frac{(9\times10^9)(2\times10^{-6} \times 4\times10^{-2})}{1\times10^{-2}\ m^2}$

B $+ 7.2 \times 10^3$ V

$V = 72 \times 10^3$

C $- 7.2 \times 10^4$ V

$V = 7.2 \times 10^4$ V

D $- 7.2 \times 10^3$ V



Topics to be covered

1. Equipotential surfaces
2. Relation between Electric Field and Potential

Very imp. topic from Numerical POV.

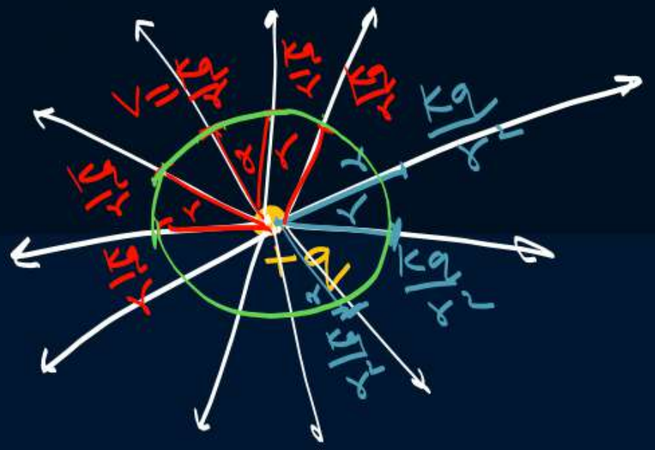
3. Electrical potential energy of a system of charges
3. Potential Energy in an External Field of a:
   1. Single charge
   2. System of two charges
   3. Dipole





[Parishram 2.0 students after last class's Dipole potential derivation]

Equipotential Surfaces

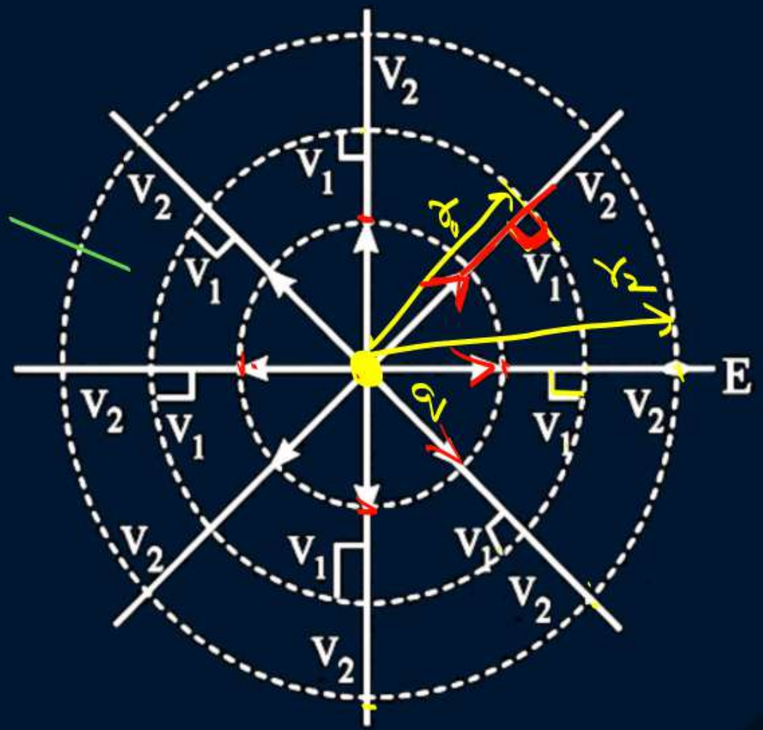


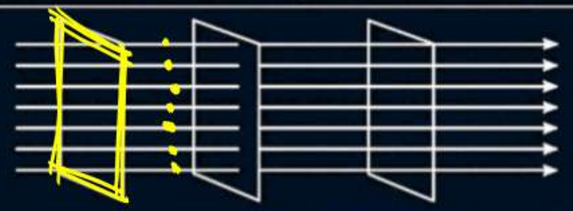
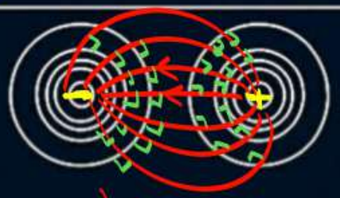
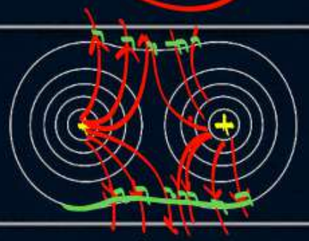
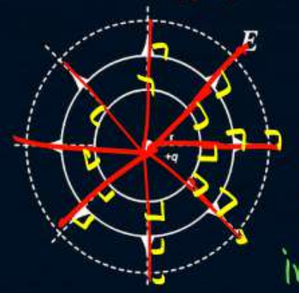
Equal

Equipotential Surfaces

These are imaginary surfaces in a region of space where electric potential is constant. They are always perpendicular to the electric field lines. They indicate regions of equal electric potential.

$\vec{E} \perp$ Equipotential surface.



System of charges	Equipotential surface	Digram
Point charge	Spherical	
Uniform electric field	Parallel planes	
Dipole	Spherical	
Identical Positive charges	Spherical	
Infinite line charge	Coaxial cylindrical	

Top View of the vertical infinite line charge.



QUESTION

The work done to move a charge on an equipotential surface is

$$\frac{W}{q} = V_2 - V_1$$

$$W = q(V_2 - V_1)$$

Same.

$W = 0$

A Less than 1

B Greater than 1

C Zero.

D Infinity.



QUESTION

Which of the following is not the property of an equipotential surface?

A They do not cross each other.

B The work done in carrying a charge from one point to another on an equipotential surface is zero.

C For a uniform electric field, they are concentric spheres.

D They are imaginary spheres.



Relation between Electric Field and Potential

$$E = \frac{kq}{r^2}$$

$\times(r)$

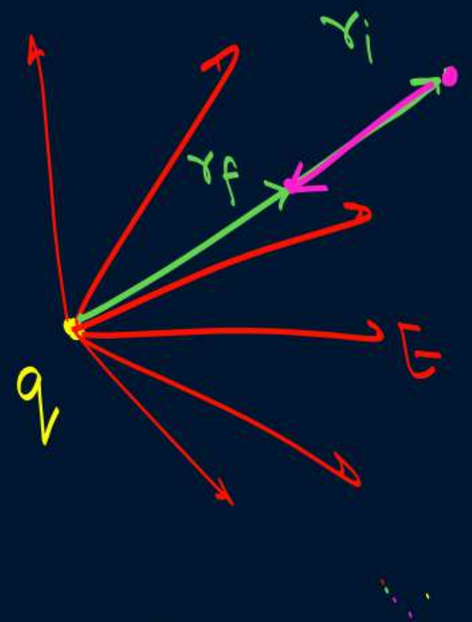
$$V = \frac{kq}{r^2} \times r$$

$$\Rightarrow V = E \times r$$

Rough Relation.

$$V = -\int_{r=r_i}^{r=r_f} \vec{E} \cdot \vec{dr}$$

Dot Product.





Relationship between Electric Field and Potential :-

$$\vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$$

$$\Rightarrow \boxed{\vec{E} = -\left(\frac{dV}{dx}\hat{i} + \frac{dV}{dy}\hat{j} + \frac{dV}{dz}\hat{k}\right)}$$

$$V = -\int \vec{E} \cdot \vec{dr}$$

$$\Rightarrow \int dV = -\int \vec{E} \cdot \vec{dr}$$

$$\Rightarrow dV = -\vec{E} \cdot \vec{dr}$$

$$\Rightarrow \boxed{\vec{E} = -\frac{dV}{\vec{dr}}}$$

The -ve sign in this expression states that as we move in the direction of Electric Field, the Electric Potential decreases.

$dx \leftarrow$ X-direc^n

$dy \leftarrow$ Y-direc^n

$dz \leftarrow$ Z-direc^n

QUESTION

In a region of constant potential

ex: $V = 5x$ V

$$E = -\frac{dV}{dx} = -\frac{d(5x)}{dx} = -5 \text{ N/C}$$

ex: $V = 5$ V

$$E = -\frac{dV}{dx} = -\frac{d(5)}{dx} = 0$$

A the electric field is uniform

B the electric field is zero

C there can be no charge inside the region.

D the electric field shall necessarily change if a charge is placed outside the region.



Electric Potential Energy of a System of Charges

Denoted by U.

S.I. unit ⇒ Joule (J)

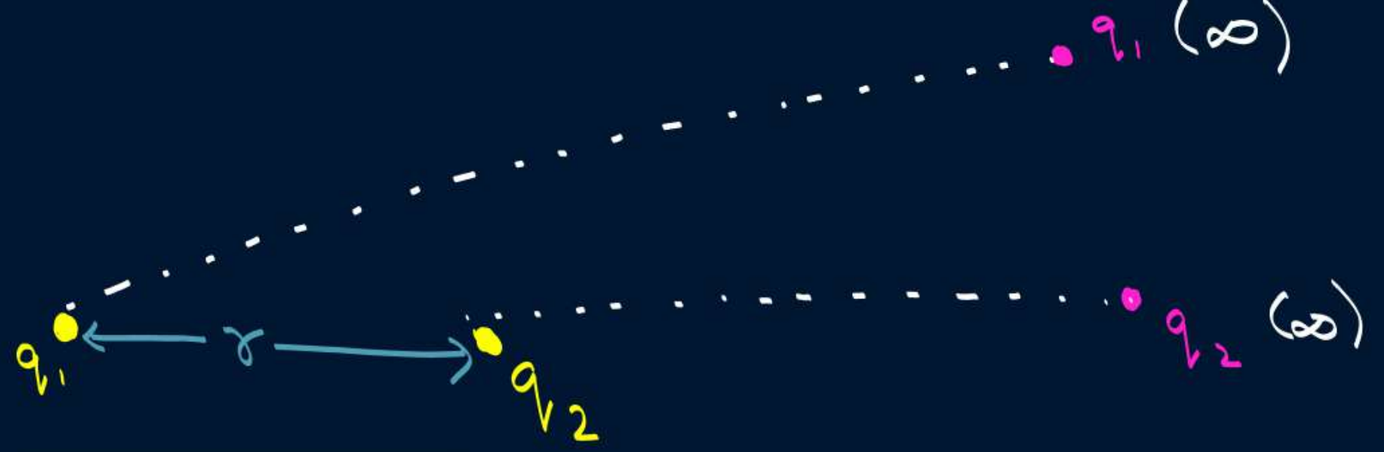
The work done in assembling a collection of charges by bringing them from infinity. It determines the stability and interactions within the system of charges.

$$W = \int_{\infty}^{r} \frac{-kq_1q_2}{r^2}\,dr$$

$$W = -kq_1q_2\left[\frac{-1}{r}\right]_{\infty}^{r}$$

$$W = \frac{kq_1q_2}{r}$$

$$U = \frac{kq_1q_2}{r}$$



→ U is a Scalar quantity.



imp.

QUESTION

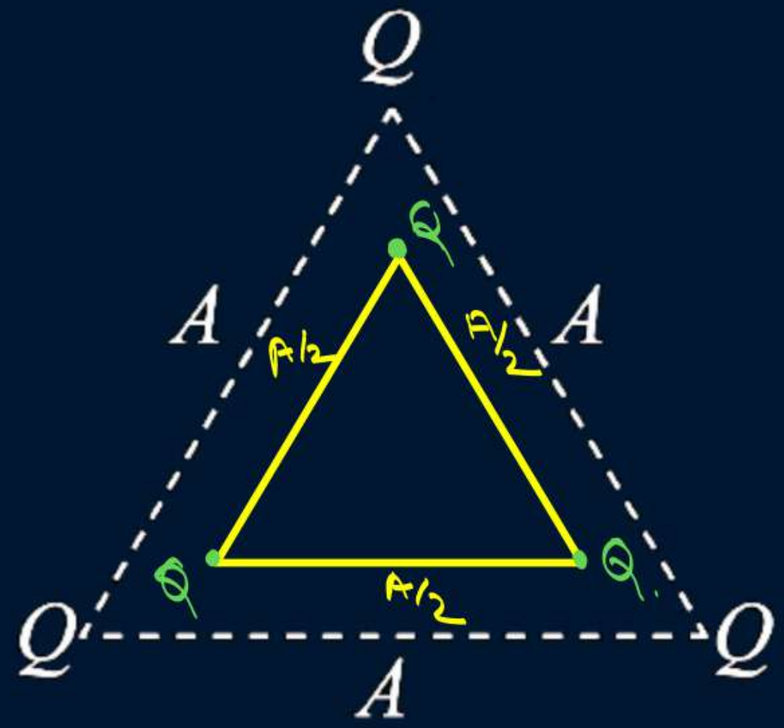
Three equal charges Q are at the vertices of an equilateral triangle of side A. How much work is done (by an external agent) in bringing them closer to an equilateral triangle of side A/2 ?

By Work Energy Theorem, we know that :-

$$W = U_f - U_i$$

$$\Rightarrow W = 3\left(\frac{kQQ}{(A/2)}\right) - 3\left(\frac{kQQ}{A}\right)$$

$$\Rightarrow W = \frac{kQ^2}{A}(6-3) \Rightarrow \boxed{W = \frac{3kQ^2}{A}}$$



QUESTION

Two point charges of + 1.0 μC are kept stationary 2 m apart. How much work is needed to be done to bring them 1 m apart?



A 4.5 mJ

B 9 mJ

C 45 mJ

D 90 mJ

$$W = U_f - U_i$$

$$W = \frac{k(1\times10^{-6})(1\times10^{-6})}{1m} - \frac{k(1\times10^{-6})(1\times10^{-6})}{2m}$$

$$W = (9\times10^{9})\times10^{-12}\left(1-\frac{1}{2}\right)$$

$$W = \frac{9}{2}\times10^{-3} = 4.5\times10^{-3}\ J$$



QUESTION

An electric field $\vec{E} = (20\hat{i} + 30\hat{j})$N/C exists in the space. If the potential at the origin is taken to be zero, find the potential at (2 m, 2 m).

$\vec{E} = (20\hat{i} + 30\hat{j})$ N/C

V_0 = Zero

$\vec{r} = (2\hat{i} + 2\hat{j})$ m

$$V = -\int \vec{E} \cdot d\vec{r}$$

$$V = -(\vec{E} \cdot \vec{r})$$

$$V = -[(20\hat{i} + 30\hat{j}) \cdot (2\hat{i} + 2\hat{j})]$$

$$V = -(40 + 60)$$

Ans. → $V = -100$ V



HW

Que ⇒ $V(r) = (2x^3\,\hat{i} + 4y\,\hat{j} + z^2\,\hat{k})$ Volts.

Find the Electric Field at the point $(1, 1, 1)$.

THANK YOU

